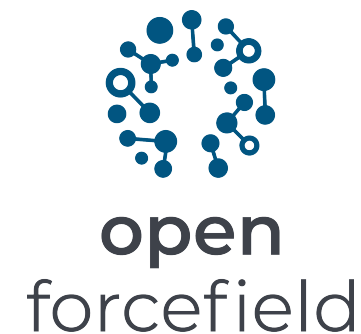


Force Field Parameterisation for Protein-Ligand Molecular Dynamics Simulations

Daniel Cole

School of Natural and Environmental Sciences



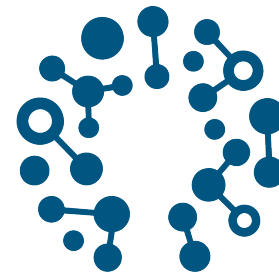
Acknowledgements

In person: Finlay Clark

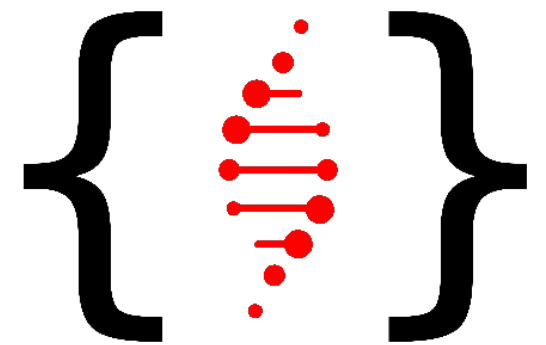
Behind the scenes:

Matt Thompson, Jeff Wagner,
Lily Wang, Ashley Mitchell,
Open Force Field

Organisers: Charlie Laughton,
James Gebbie-Rayet,
Shivani Harshe, Sarah Harris



open
forcefield



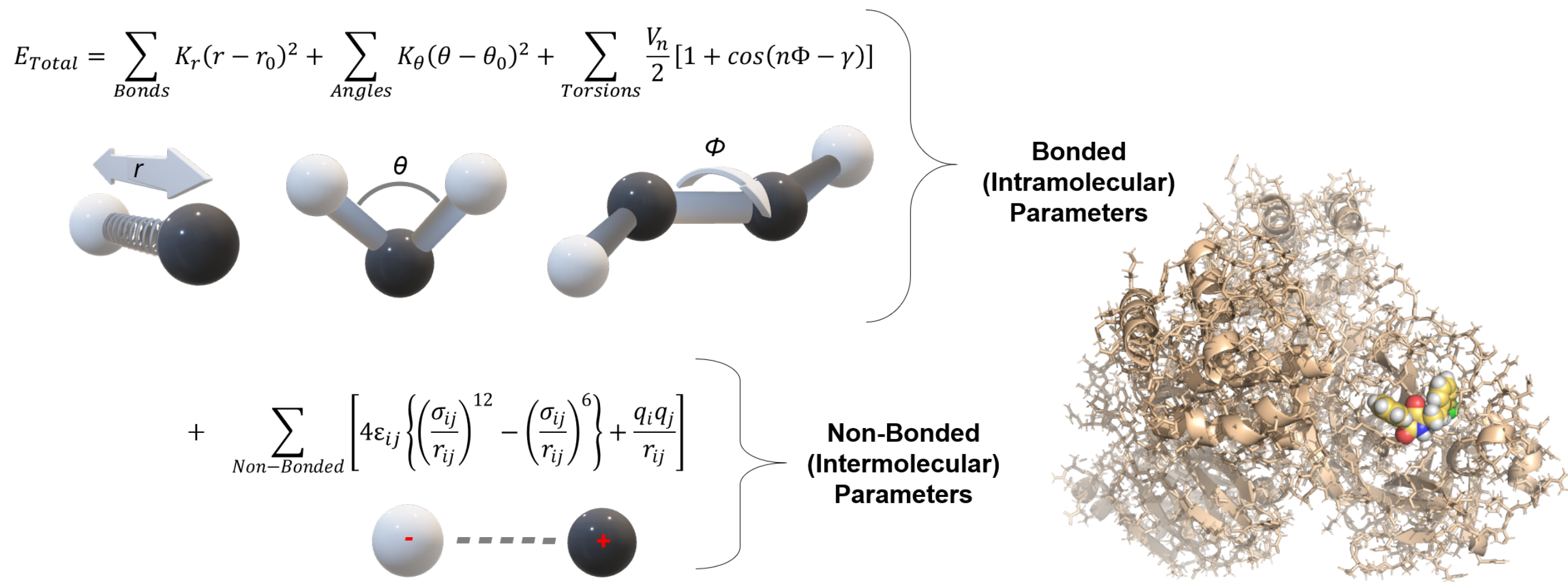
CCP BioSim



UK Research
and Innovation

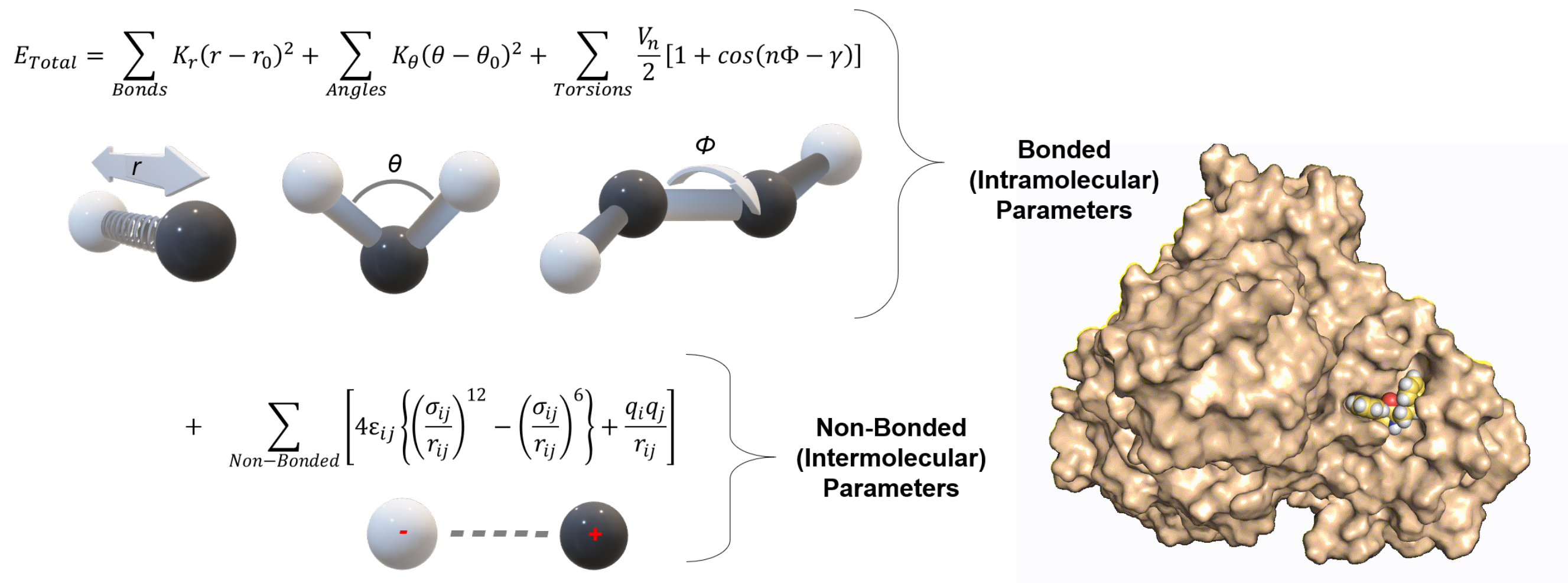
Force Field

Simulations of **large** chemical systems over **long** time scales (e.g. free energy calculations for computer-aided drug design) are crucially reliant on the accuracy of the molecular mechanics force field.



Force Field

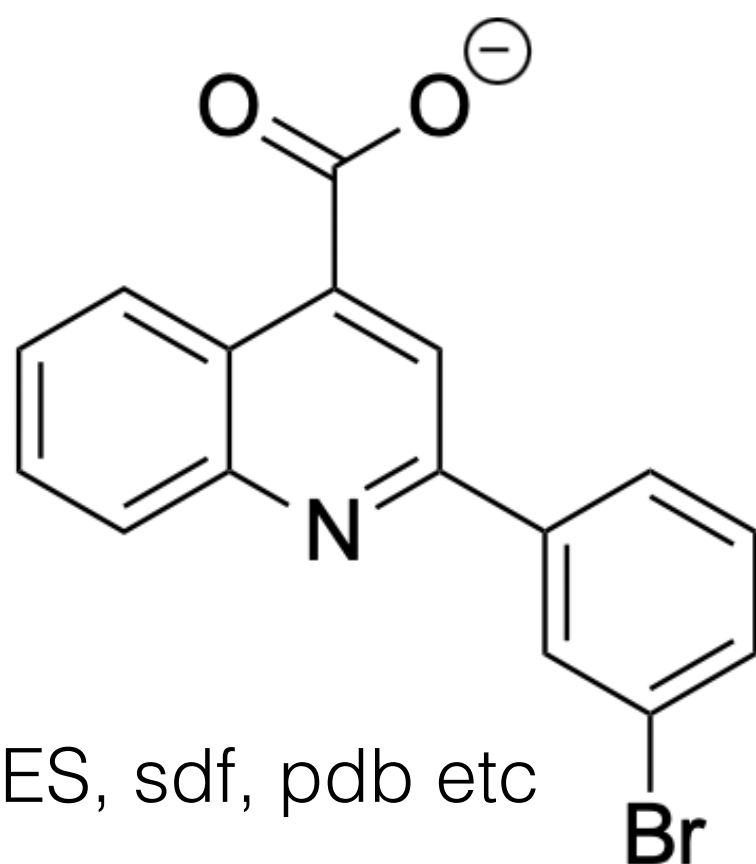
Simulations of **large** chemical systems over **long** time scales (e.g. free energy calculations for computer-aided drug design) are crucially reliant on the accuracy of the molecular mechanics force field.



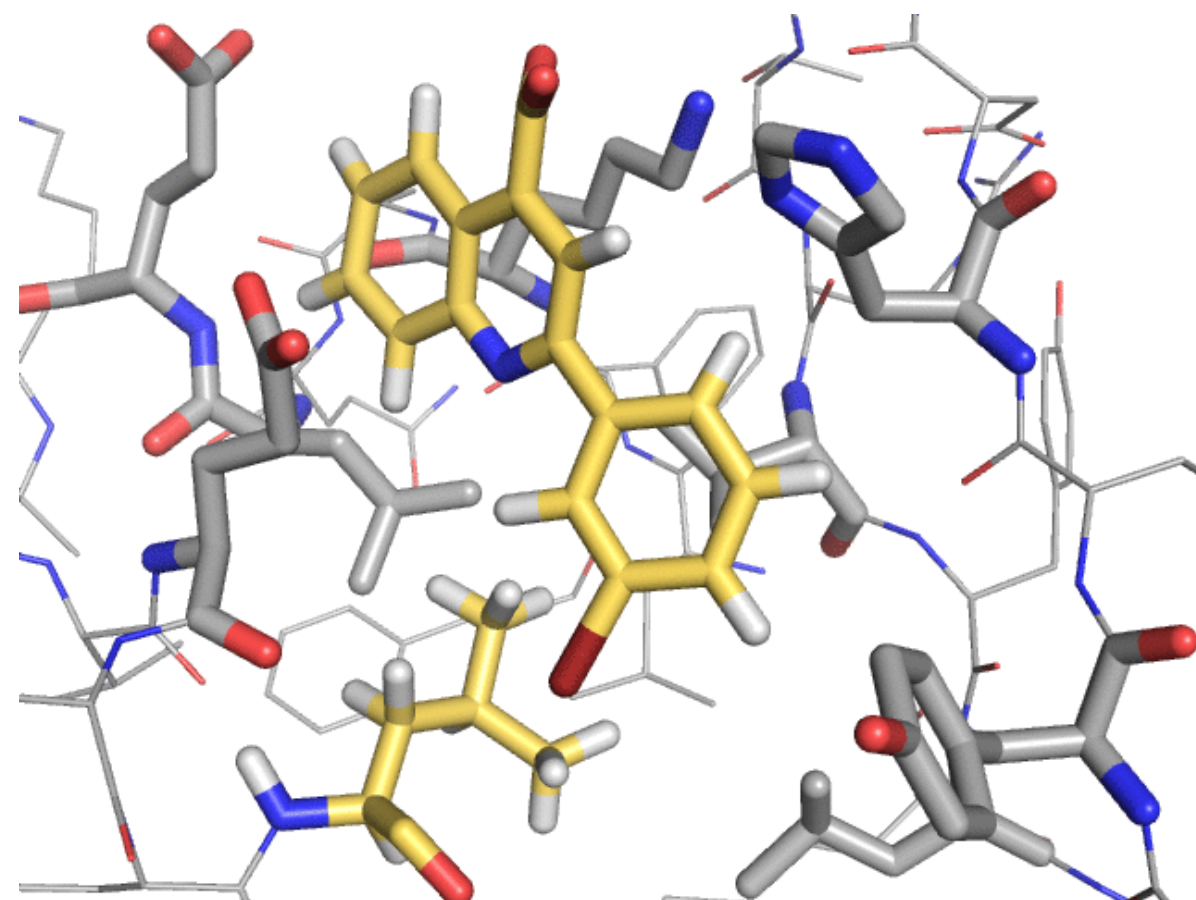
Force Field

Typically need a protein force field (e.g. AMBER, CHARMM, OPLS), a small molecule force field (e.g. GAFF, CGenFF) and a water force field (e.g. TIP3P).

Can also increasingly use machine learning potentials (e.g. ANI, MACE-OFF, Espaloma).

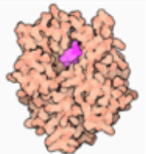
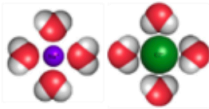
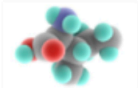
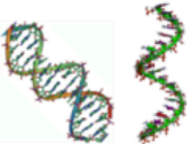
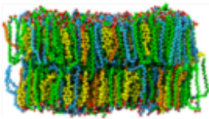
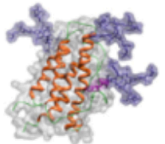


SMILES, sdf, pdb etc



Force Field

Molecular mechanics force fields have traditionally been products of heroic endeavour (> 100 human-years). How can we bring this into the modern era?

	proteins post-translational modifications
	water ions
	small molecules
	nucleic acids
	lipids
	carbohydrates

Amber20 recommendations

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Open Force Field

The Open Force Field Initiative is an international academic-industry collaboration working together to advance science & infrastructure required to build force fields.



OPEN SOFTWARE

Automated infrastructure enables rapid experimentation with minimum human intervention



OPEN DATA

Access to large, high quality experimental and quantum chemical data facilitates easy curation of balanced train / test sets

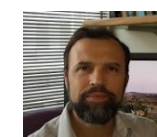


OPEN SCIENCE

Exploring new force field science:
hypothesis - build
software - train - test -
iterate
is now almost routine



John Chodera (MSKCC)



Michael Gilson (UCSD)



David Mobley (UC Irvine)



Michael Shirts (CU Boulder)



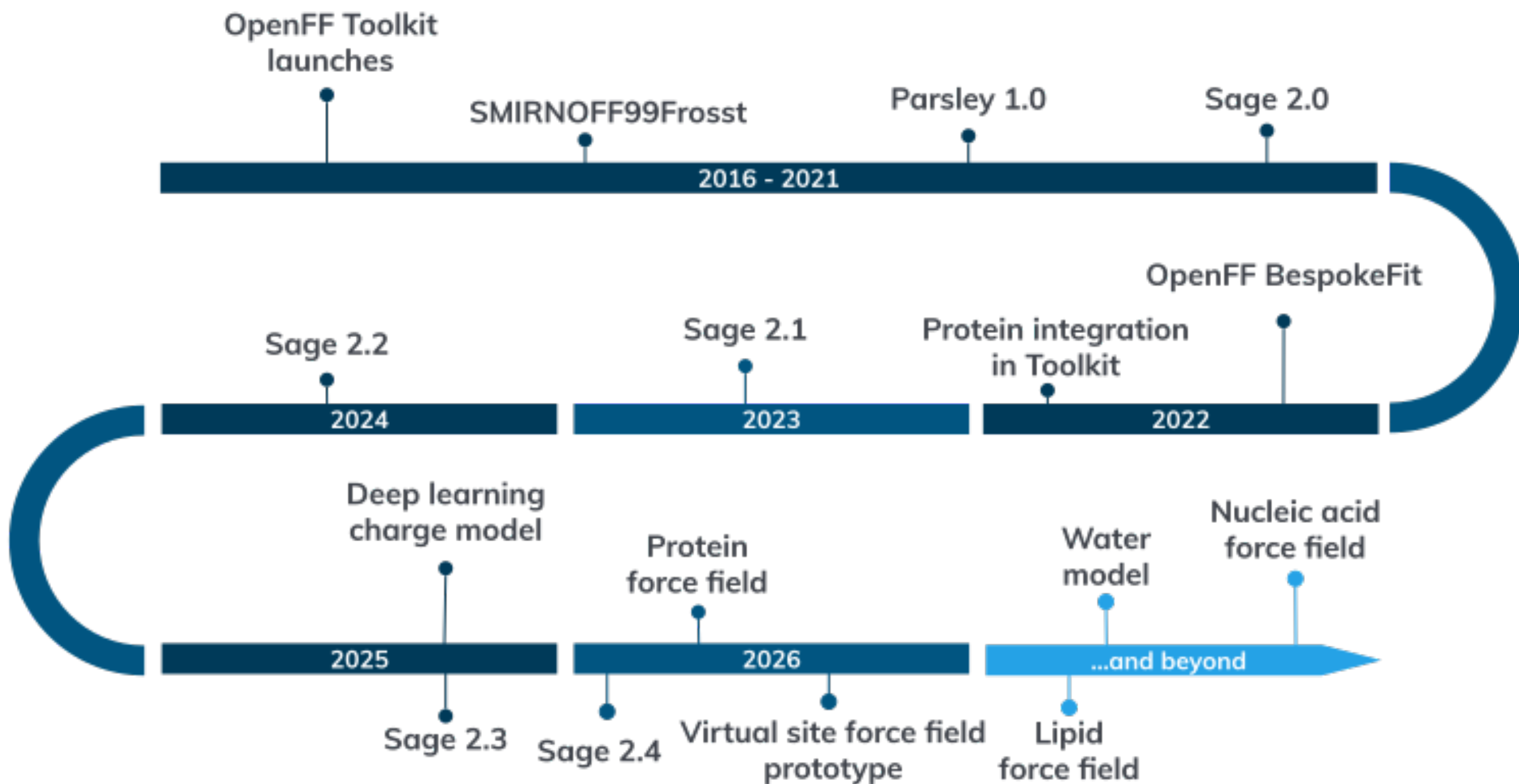
Jeff Wagner
Technical Lead



Lily Wang
Science Lead

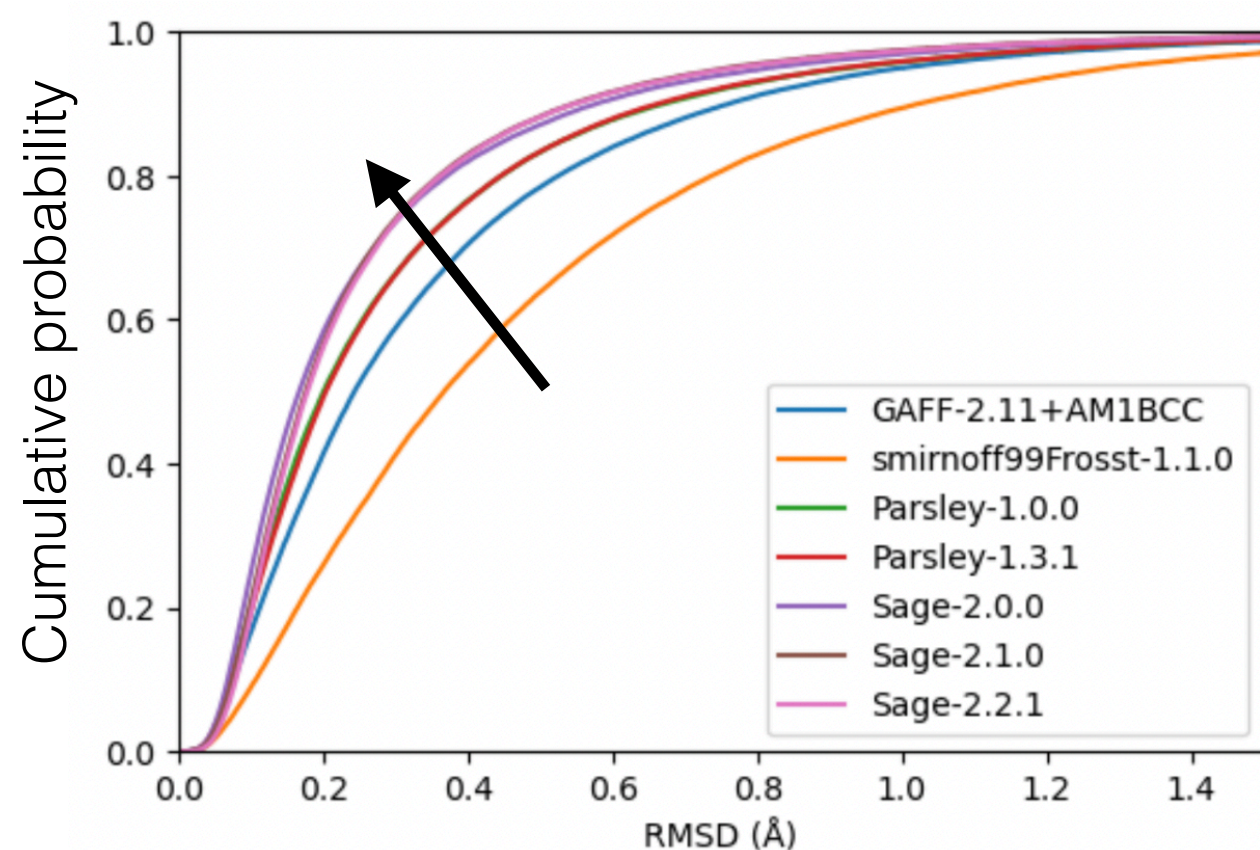
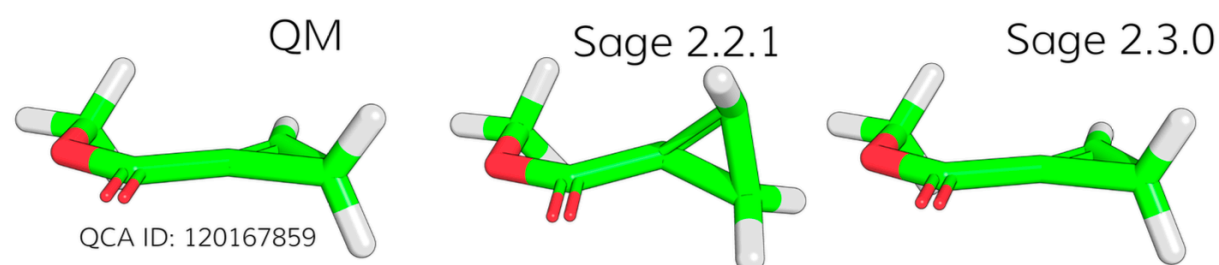
<https://openforcefield.org/>

Open Force Field Timeline



*Sage 2.3.0: Wang et al., JCTC, **2026**, 22, 4507-4531*

Sage is already a leading public small molecule force field



Parsley 1.x

Re-fit bonds, angles, torsions



Sage 2.x

Re-fit valence, LJ parameters



Sage 2.3.0 released!

with AshGC neural network charges



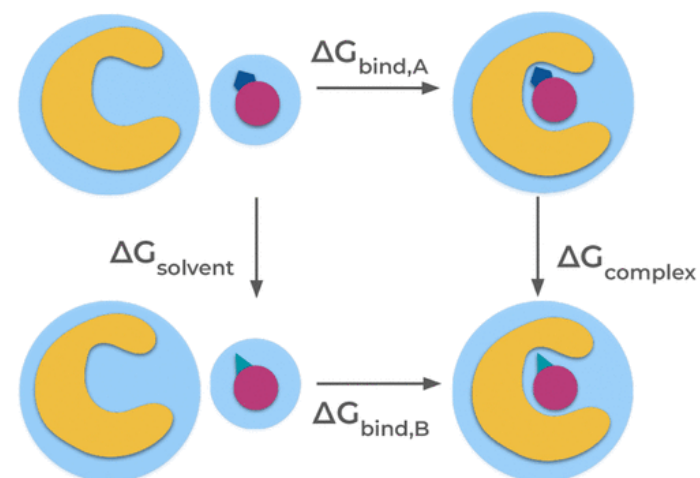
Rosemary 3.0 alpha released

Preliminary peptide and small molecule parameters

Rosemary 3.0

Expansion to protein parameters

Sage is already a leading public small molecule force field



...see tomorrow



Parsley 1.x

Re-fit bonds, angles, torsions



Sage 2.x

Re-fit valence, LJ parameters



Sage 2.3.0 released!

with AshGC neural network charges

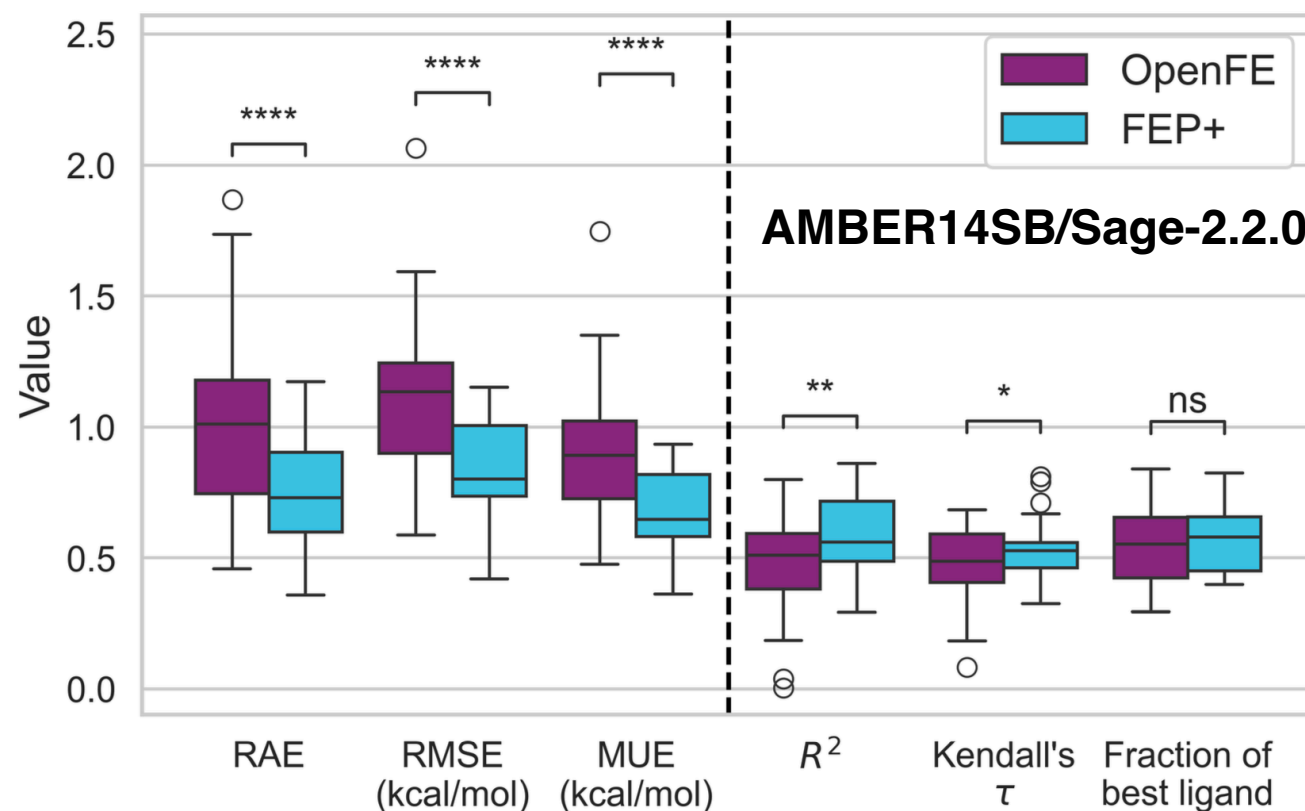


Rosemary 3.0 alpha released

Preliminary peptide and small molecule parameters

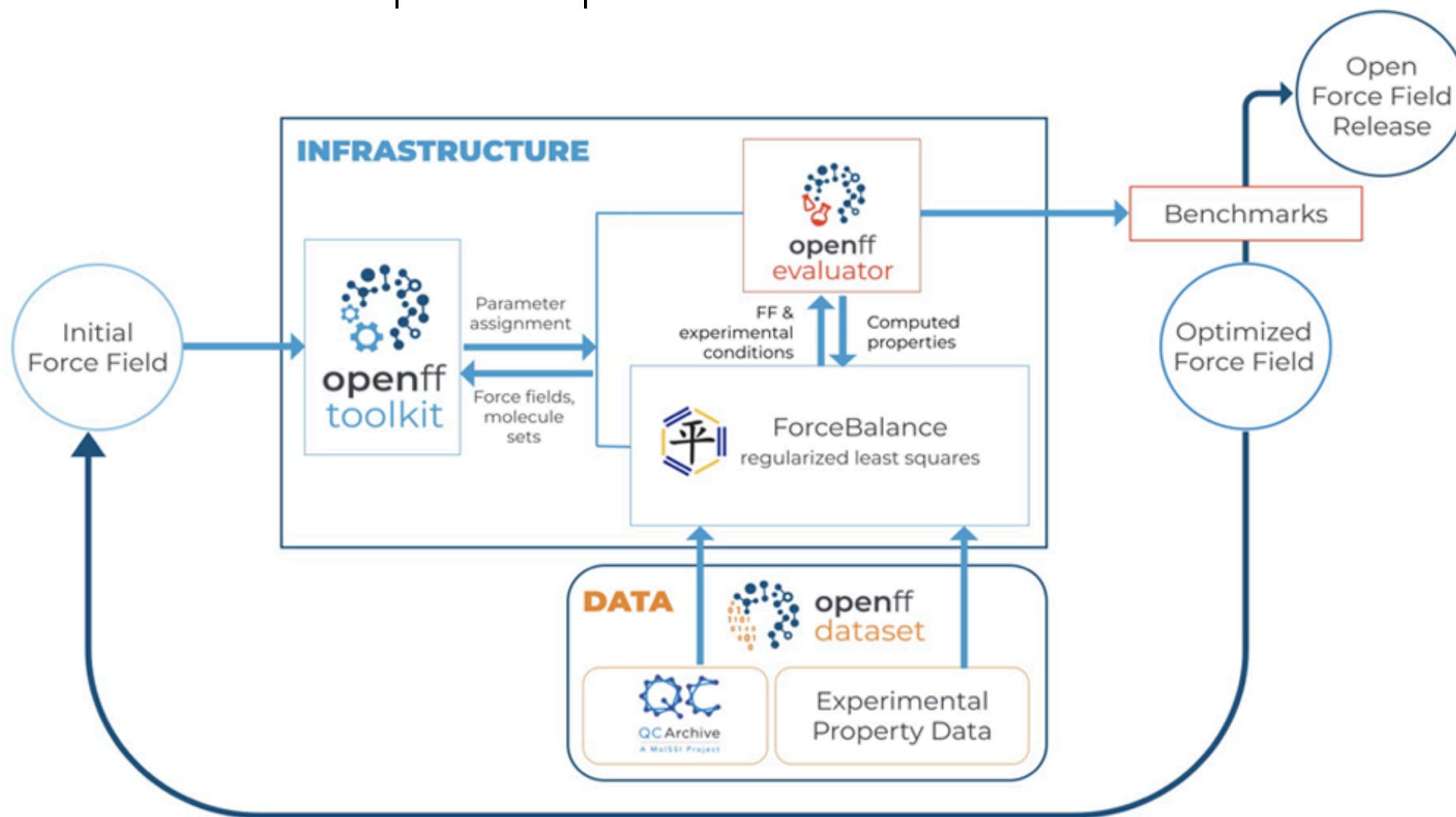
Rosemary 3.0

Expansion to protein parameters



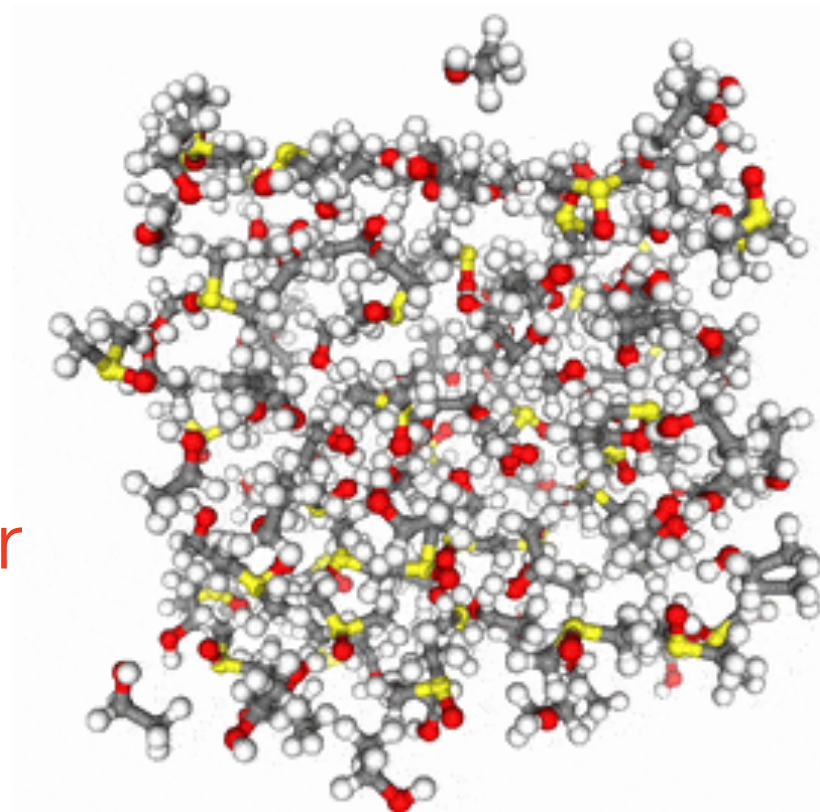
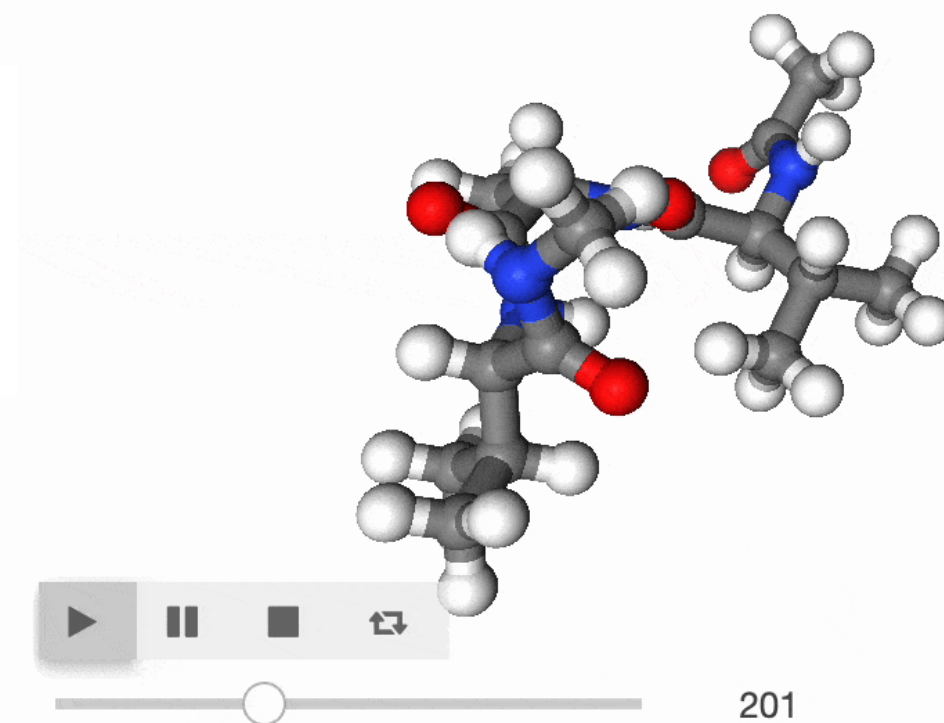
Open Force Field

Open Force Field infrastructure significantly reduces human time required to parameterise a new force field.



Training Data

Bonded parameters trained on curated QM energies and forces of 1000's of drug-like molecules and fragments

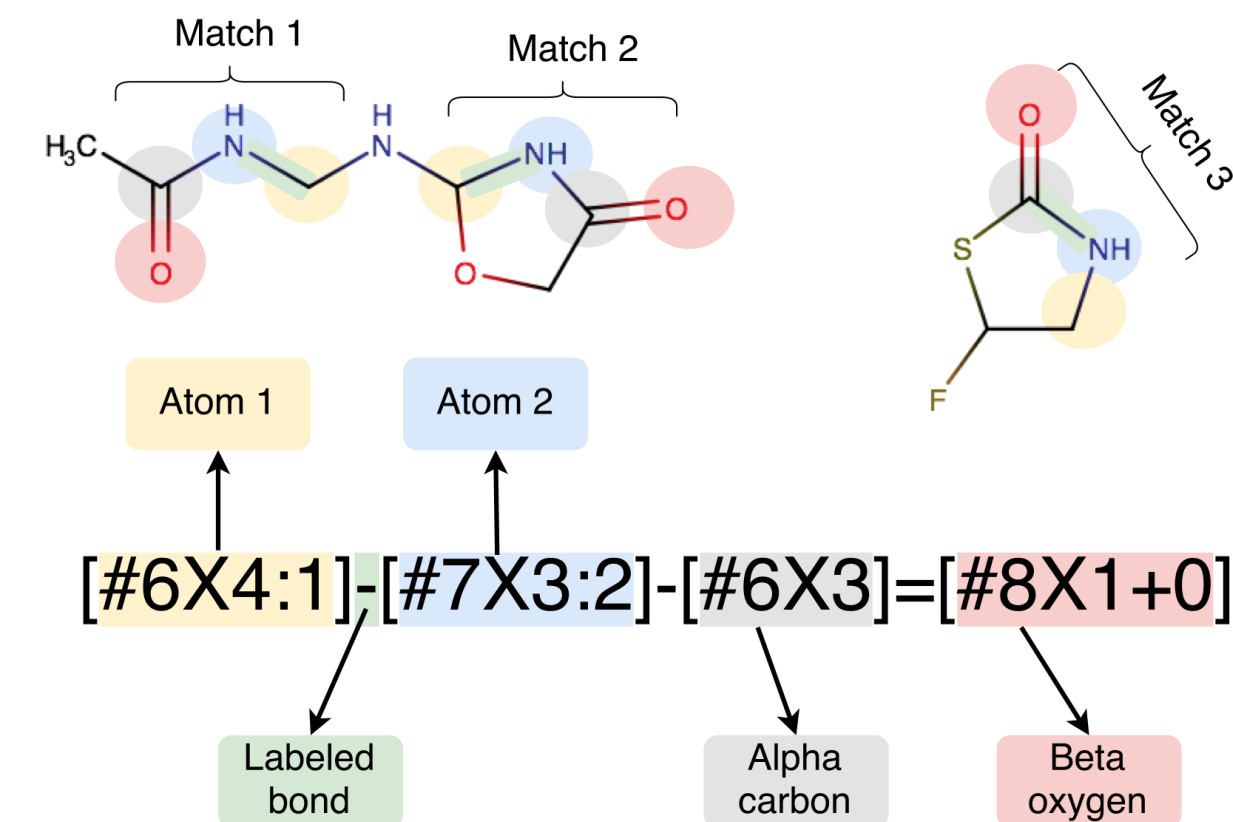


Non-bonded training set includes 1000's of mixture enthalpy and density measurements from NIST ThermoML

SMIRKS Native Open Force Field Specification

Direct chemical perception bypasses atom typing and assigns parameters to atoms/bonds/angles/torsions by processing the full molecular graph.

Mobley et al., JCTC, 2018, **14**, 6076



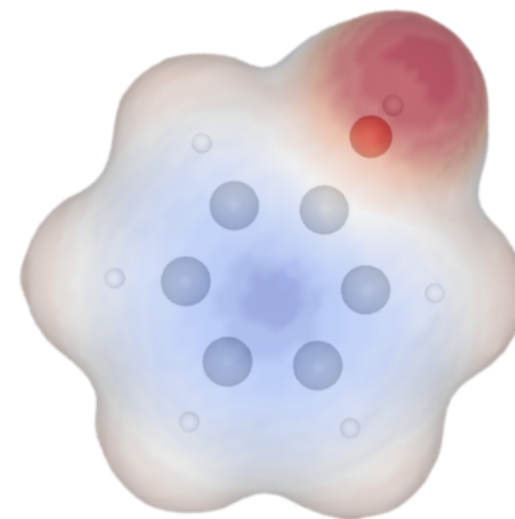
Can avoid a proliferation of parameters without loss of accuracy.

parameter type	MMFF ⁵	OPLS3 ⁵	OPLS3e ⁴	Sage (OpenFF 2.0.0) ⁶
bond stretching	456	1187	1187	88
angle bending	2283	15,236	15,235	40
torsional rotation	520	48,142	146,669	167

Atomic Charges

Most parameters are read in from a library, but exceptions are the atomic charges.

We typically want the charges to reproduce the electrostatic potential (ESP) around the molecule while accounting for condensed phase polarisation.

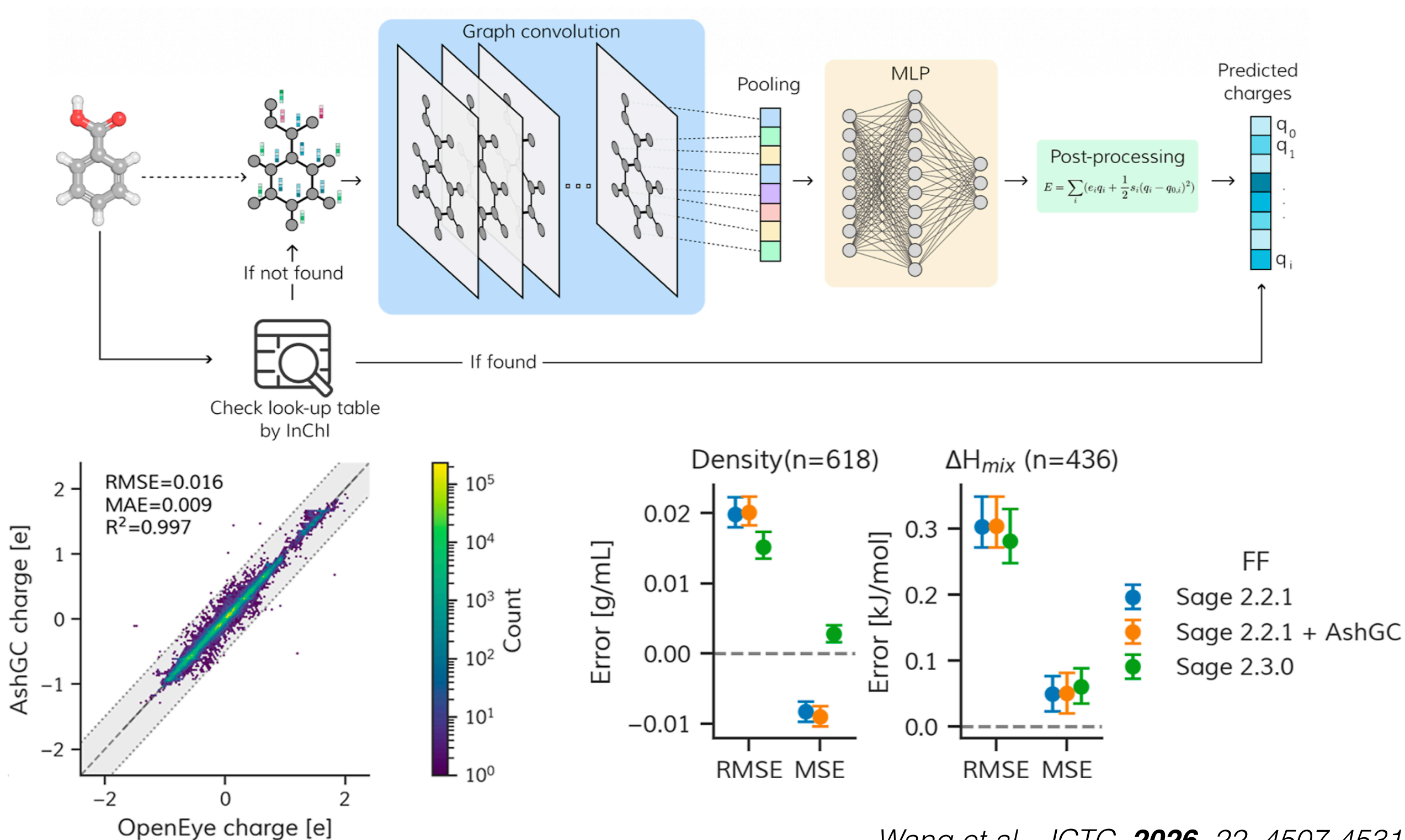


Example charge model is RESP, fit to HF/6-31G(d) QM.

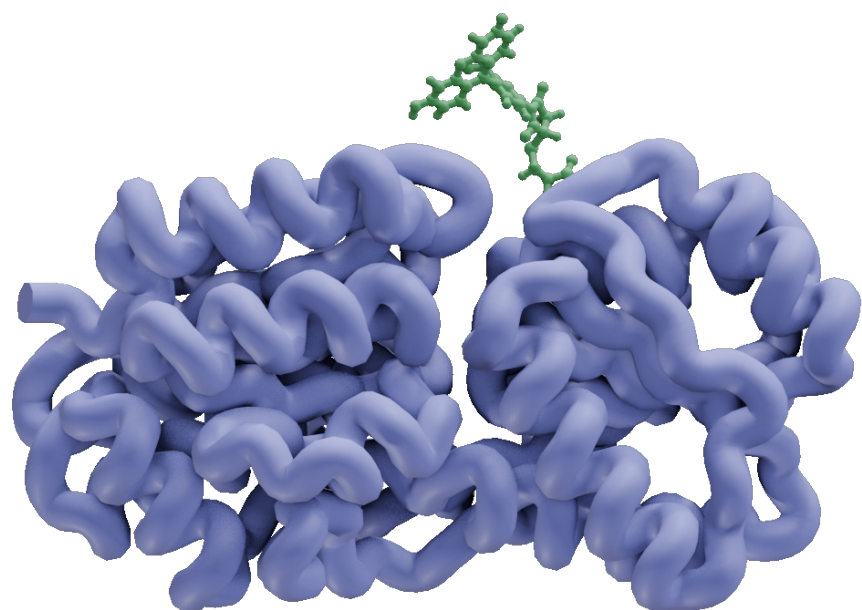
This can be expensive for high-throughput calculations, so alternative is AM1-BCC based on semi-empirical calculations (method used in Parsley/Sage).

This can still be expensive! So a new alternative is a graph-based charge model (AshGC), which is the default charge method in Sage 2.3.0.

AshGC: same accuracy for fraction of cost



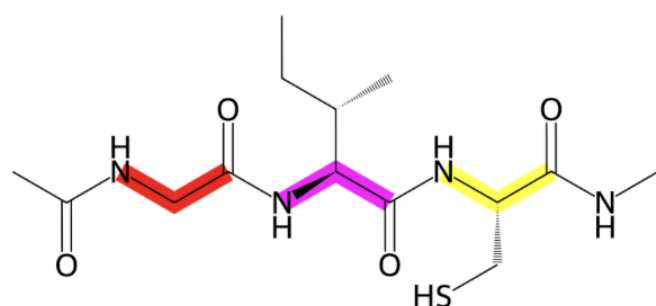
Rosemary (alpha): small molecule and protein force field



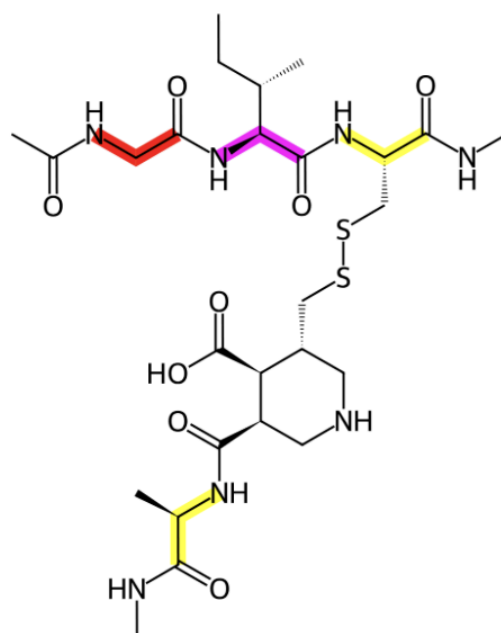
AshGC enables fast assignment of charges to both small molecules and proteins



Chapin Cavender, UCSD



GLY-ILE-CYS



+ covalent ligand

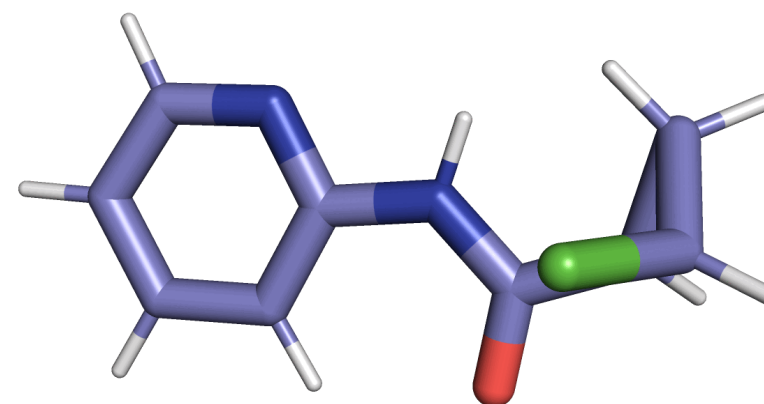
Rosemary (alpha) is just Sage with 6 more torsions. Parameters are co-trained to both small molecule and peptide data.

New torsions (phi+psi): **Glycine** **Beta-branched** **Sidechain**

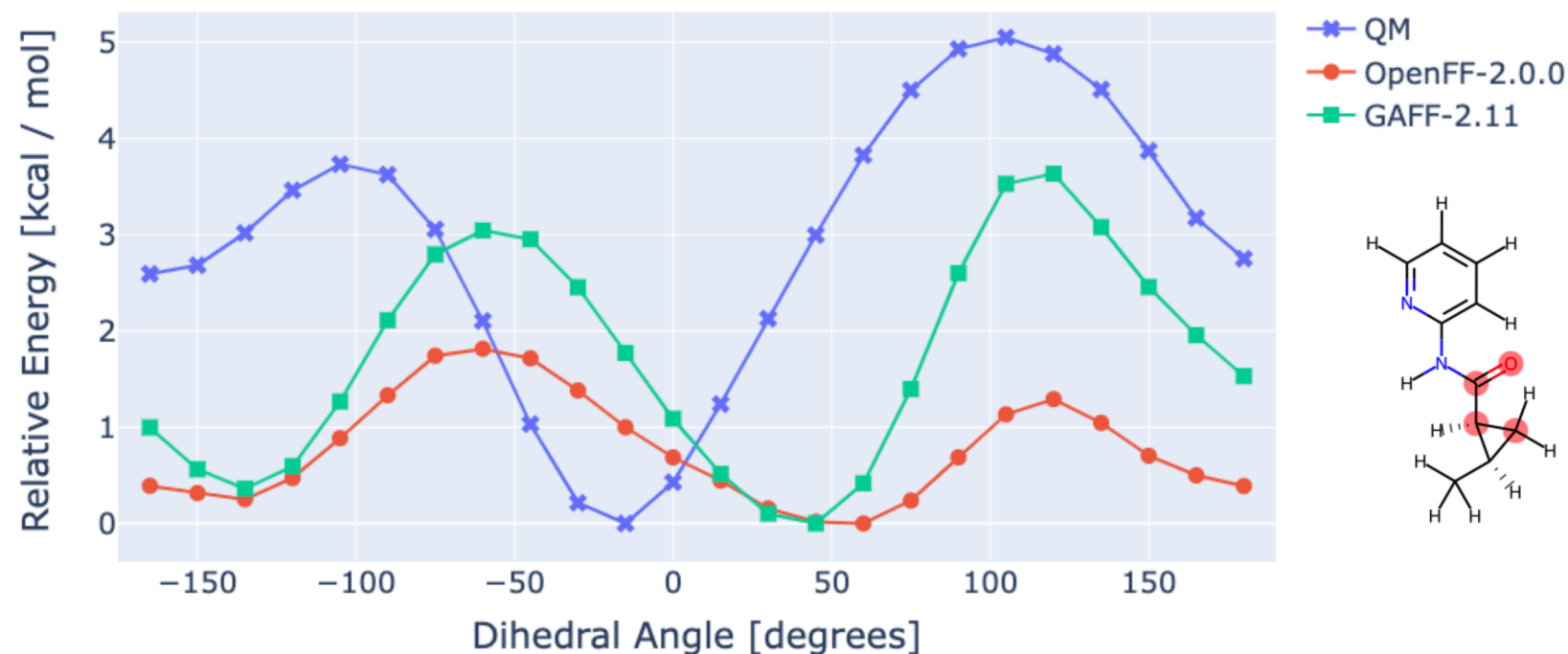
Bespoke Force Field Parameters

Accurate determination of molecular conformation is crucial in structure-based drug design.

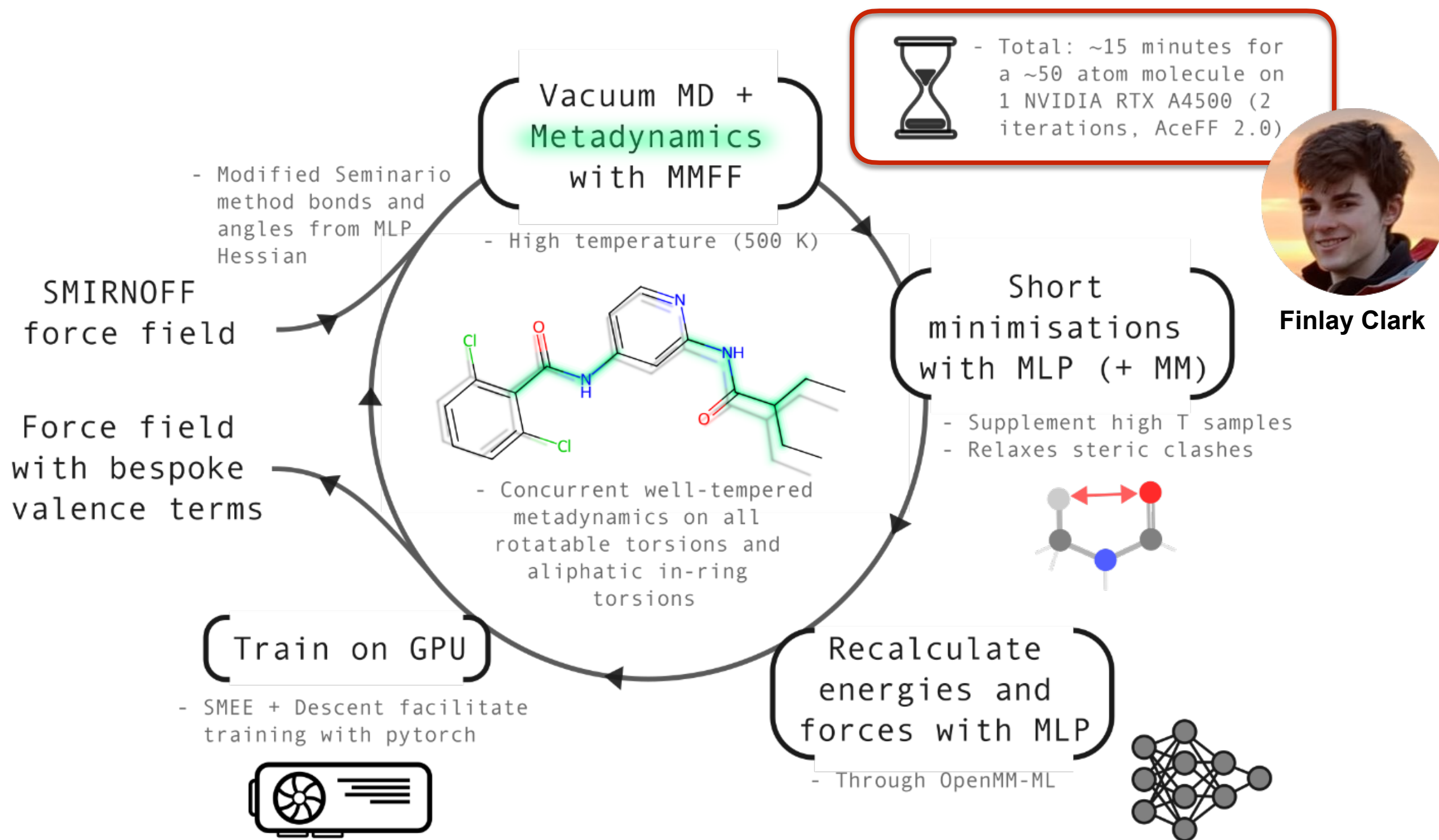
Molecular conformation is largely determined by torsional rotation about flexible bonds.



Transferability is difficult due to sensitivity to surrounding environment.



Presto: GPU-Accelerated Bespoke Fitting



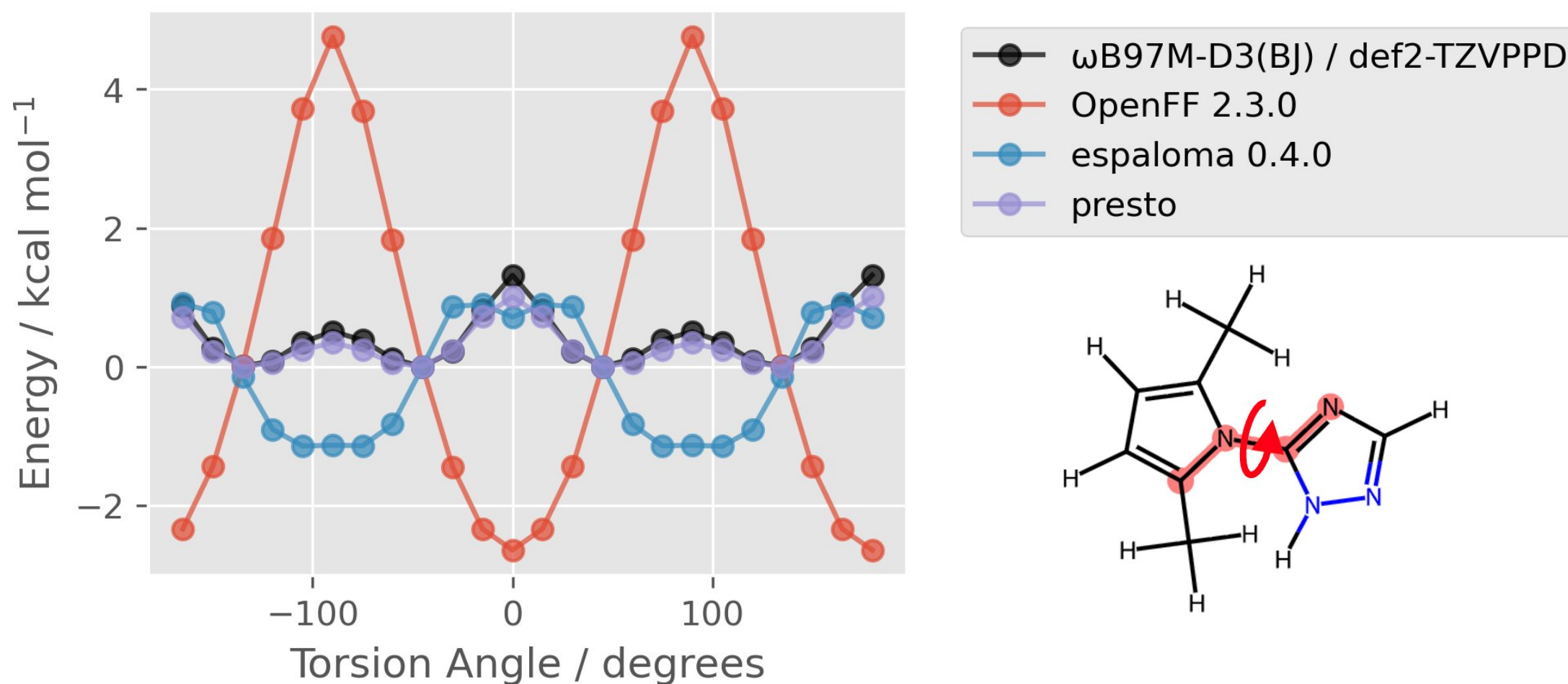
<https://github.com/cole-group/presto>

Finlay Clark et al. (2026), <https://doi.org/10.26434/chemrxiv.15004169/v2>

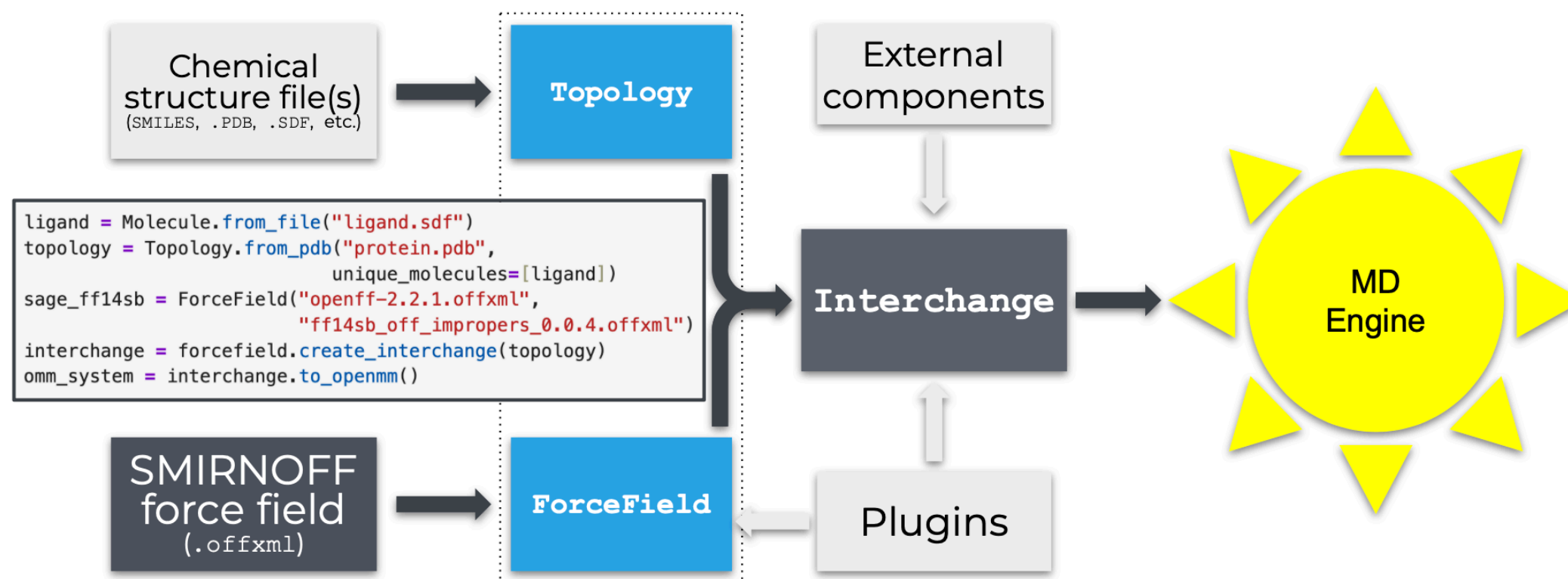
Torsion Scans

Force fields are easily exported for use in simulation or the OpenFF benchmarking suite.

Torsion scan statistics on 400 fragments from TorsionNet500 (Presto trained to AceFF 2.0, no explicit torsion scans).



Today's Workshops



Finlay Clark

The OpenFF workflow lets you set up MD simulations in a few lines of code

2.30-4pm: Ligand parameterisation and charge assignment (including AshGC)

4-5pm: Running MD and analysis of a protein-ligand system

The protein target

Journal of Medicinal Chemistry > Vol 56/Issue 1 > Article

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FEATURED ARTICLE | December 17, 2012

Discovery of Potent Myeloid Cell Leukemia 1 (Mcl-1) Inhibitors Using Fragment-Based Methods and Structure-Based Design

Anders Friberg, Dominico Vigil, Bin Zhao, R. Nathan Daniels, Jason P. Burke, Pedro M. Garcia-Barrantes, DeMarco Camper, Brian A. Chauder, Taekyu Lee, Edward T. Olejniczak, and Stephen W. Fesik*

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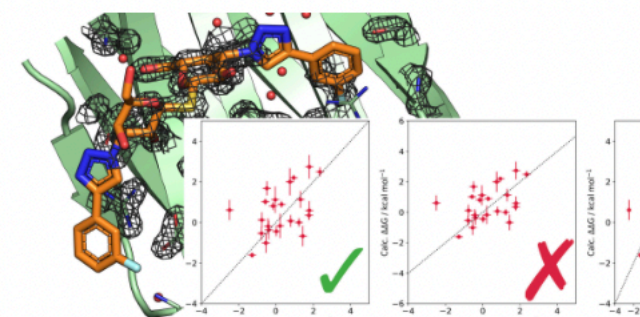
Best Practices for Constructing, Preparing, and Evaluating Protein-Ligand Binding Affinity Benchmarks [Article v1.0]

David F. Hahn

Computational Chemistry, Janssen Research & Development, Turnhoutseweg 30,
Beerse B-2340, Belgium

<https://orcid.org/0000-0003-2830-6880>

Christopher I. Bayly



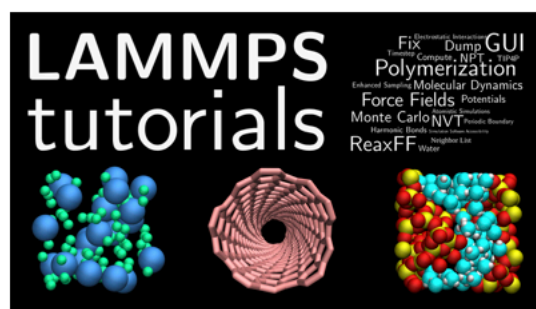
Where can I find out more?

Best Practices for Foundations in Molecular Simulations [Article v1.0]

Efrem Braun¹, Justin Gilmer², Heather B. Mayes³, David L. Mobley⁴, Jacob I. Monroe⁵, Samarjeet Prasad⁶, Daniel M. Zuckerman⁷

A suite of tutorials for the BioSimSpace framework for interoperable biomolecular simulation [Article v1.0]

Lester O. Hedges^{1,2*}, Sofia Bariami^{3†}, Matthew Burman², Finlay Clark³, Benjamin P. Cossins⁴, Adele Hardie³, Anna M. Herz³, Dominykas Lukauskis⁵, Antonia S.J.S. Mey³, Julien Michel^{2,3*}, Jenke Scheen^{3‡}, Miroslav Suruzhon⁴, Christopher J. Woods¹, Zhiyi Wu⁴



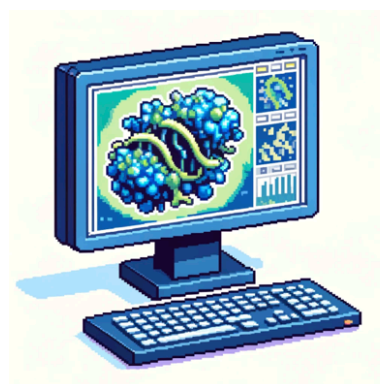
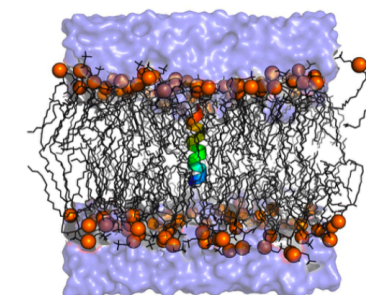
From Proteins to Perturbed Hamiltonians: A Suite of Tutorials for the GROMACS-2018 Molecular Simulation Package [Article v1.0]

Justin A. Lemkul

Department of Biochemistry, Virginia Polytechnic Institute and State University
<https://orcid.org/0000-0001-6661-8653>

DOI: <https://doi.org/10.33011/livecoms.1.1.5068>

Keywords: tutorials, gromacs, molecular dynamics simulation, computational chemistry



Molecular Dynamics: From Basics to Application [Article v1.0]

Luis Vollmers, Shu-Yu Chen, Maria Reif, Tristan Alexander Mauck, Martin Zacharias

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A Set of Tutorials for the LAMMPS Simulation Package [Article v1.0]

Simon Gravelle, Cecilia M. S. Alvares, Jacob R. Gissinger, Axel Kohlmeyer

[Article Code Repository](#)

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<https://docs.openforcefield.org/en/latest/examples.html>

A background image showing a molecular structure with red and white spheres connected by grey rods, set against a dark blue, textured background.

Living Journal of Computational Molecular Science

Living Journal of Computational Molecular Science is a diamond open access journal sharing best practices in molecular simulation.

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